N B
Study Material
(Operating System)

Module I

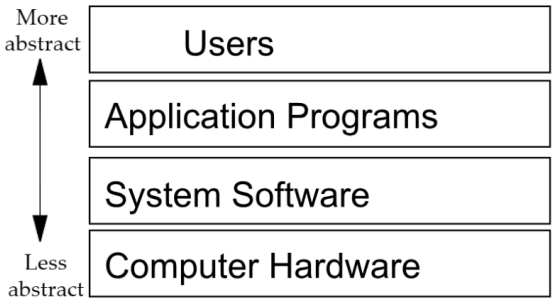
Table of Contents:

1. Basic OS functions, resource abstraction, types of operating systems–multiprogramming systems, batch systems, time sharing systems.
2.Operating systems for personal computers & workstations.
3. Process control & real time systems.
4. Processor and user modes, kernels, system calls and system programs.

1. Basic OS functions

Operating System

An operating system is a program on which application programs are executed and acts as a communication bridge (interface) between the user and the computer hardware. The main task an operating system carries out is the allocation of resources and services, such as the allocation of memory, devices, processors, and information. The operating system also includes programs to manage these resources, such as a traffic controller, a scheduler, a memory management module, I/O programs, and a file system.



Why are Operating Systems Used?

Operating System is used as a communication channel between the Computer hardware and the user. It works as an intermediate between System Hardware and End-User. Operating System handles the following responsibilities:

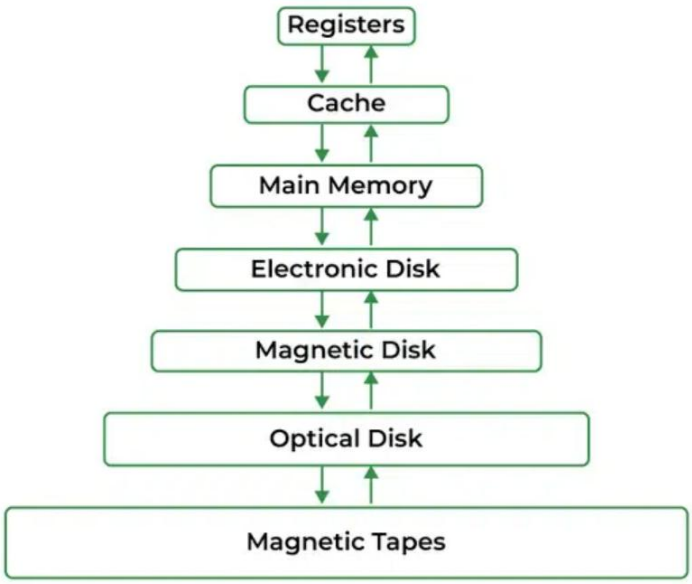
- It controls all the computer resources.
- It provides valuable services to user programs.
- It coordinates the execution of user programs.
- It provides resources for user programs.
- It provides an interface (virtual machine) to the user.
- It hides the complexity of software.
- It supports multiple execution modes.
- It monitors the execution of user programs to prevent errors.

Functions of an Operating System

Memory Management

The operating system manages the Primary Memory or Main Memory. Main memory is made up of a large array of bytes or words where each byte or word is assigned a certain address. Main memory is fast storage and it can be accessed directly by the CPU. For a program to be executed, it should be first loaded in the main memory. An operating system manages the allocation and deallocation of the memory to various processes and ensures that the other process does not consume the memory allocated to one process. An Operating System performs the following activities for Memory Management:

- It keeps track of primary memory, i.e., which bytes of memory are used by which user program. The memory addresses that have already been allocated and the memory addresses of the memory that has not yet been used.
- In multiprogramming, the OS decides the order in which processes are granted memory access, and for how long.
- It Allocates the memory to a process when the process requests it and deallocates the memory when the process has terminated or is performing an I/O operation.



Resource abstruction

Resource abstraction is a concept in computer science and software engineering that refers to the process of hiding implementation details and complexities of accessing and using system resources, services, or components behind a simplified and consistent interface. The goal of resource abstraction is to provide a higher-level and more user-friendly representation of the underlying resources, allowing developers to interact with them in a more straightforward and standardized manner.

Resource abstraction is commonly used in various software design patterns and architectures to achieve the following benefits:

1. Simplification: By abstracting away the low-level details, developers can focus on the core functionality of their applications without getting bogged down in the intricacies of the underlying resources.
2. Reusability: Abstracting resources allows developers to create a single, uniform interface that can be used to interact with different underlying implementations. This promotes code reusability and makes it easier to switch or extend the underlying resources without affecting the rest of the application.
3. Encapsulation: Abstraction encapsulates the internal workings of the resources, reducing dependencies and potential points of failure in the codebase.
4. Consistency: A well-designed resource abstraction provides a consistent and predictable interface for working with various resources, making it easier for developers to learn and use different parts of the system.

Common examples of resource abstraction include:

1. File System Abstraction: Instead of directly interacting with the operating system's file system functions, a programming language might provide an abstraction layer (e.g., the **File** class) that allows developers to work with files using simple methods like **read**, **write**, and **delete**.
2. Database Abstraction: Object-Relational Mapping (ORM) frameworks abstract the database interactions, allowing developers to interact with databases using high-level programming constructs rather than writing raw SQL queries.
3. Network Abstraction: Network libraries provide abstractions for sending and receiving data over networks, shielding developers from the complexities of handling low-level network protocols.
4. Memory Abstraction: Programming languages abstract memory management through features like garbage collection, enabling developers to focus on the logic of their applications without worrying about manual memory allocation and deallocation.

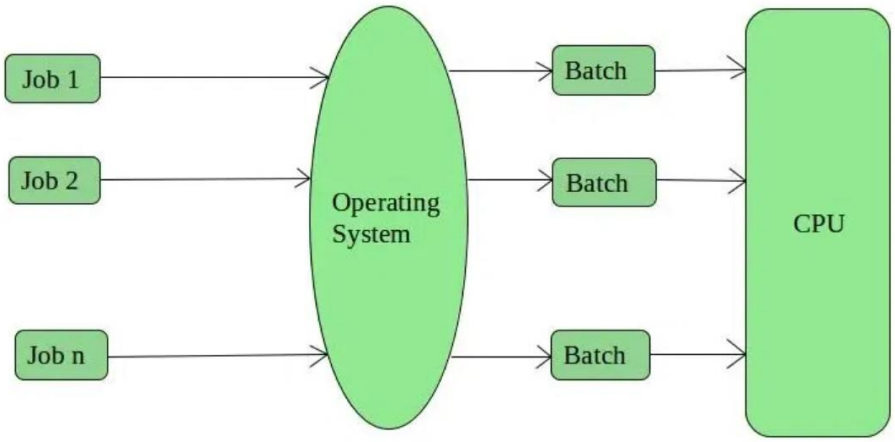
Resource abstraction is a crucial aspect of software design as it promotes modularity, maintainability, and flexibility, making it easier to build complex systems and manage their interactions with various resources.

Types of operating systems:

- Batch Operating System
- Multi-Programming System
- Multi-Processing System
- Multi-Tasking Operating System
- Time-Sharing Operating System
- Distributed Operating System
- Network Operating System
- Real-Time Operating System

1. Batch Operating System

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and groups them into batches. It is the responsibility of the operator to sort jobs with similar needs.



Advantages of Batch Operating System

- It is very difficult to guess or know the time required for any job to complete. Processors of the batch systems know how long the job would be when it is in the queue.
- Multiple users can share the batch systems.
- The idle time for the batch system is very less.
- It is easy to manage large work repeatedly in batch systems.

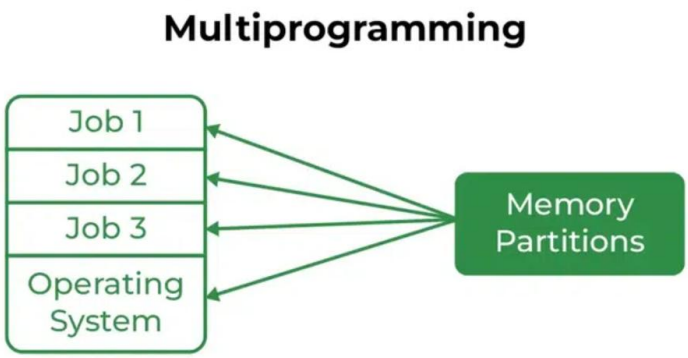
Disadvantages of Batch Operating System

- The computer operators should be well known with batch systems.
- Batch systems are hard to debug.
- It is sometimes costly.
- The other jobs will have to wait for an unknown time if any job fails.

Examples of Batch Operating Systems: Payroll Systems, Bank Statements, etc.

2. Multi-Programming Operating System

Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution. This is basically used for better execution of resources.



Advantages of Multi-Programming Operating System

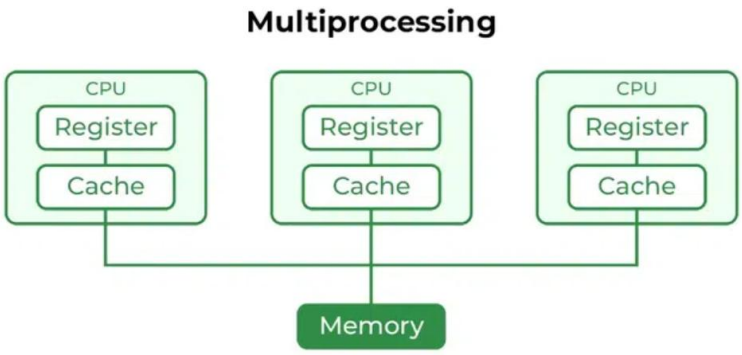
- Multi Programming increases the Throughput of the System.
- It helps in reducing the response time.

Disadvantages of Multi-Programming Operating System

- There is not any facility for user interaction of system resources with the system.

3. Multi-Processing Operating System

Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It betters the throughput of the System.



Advantages of Multi-Processing Operating System

- It increases the throughput of the system.
- As it has several processors, so, if one processor fails, we can proceed with another processor.

Disadvantages of Multi-Processing Operating System

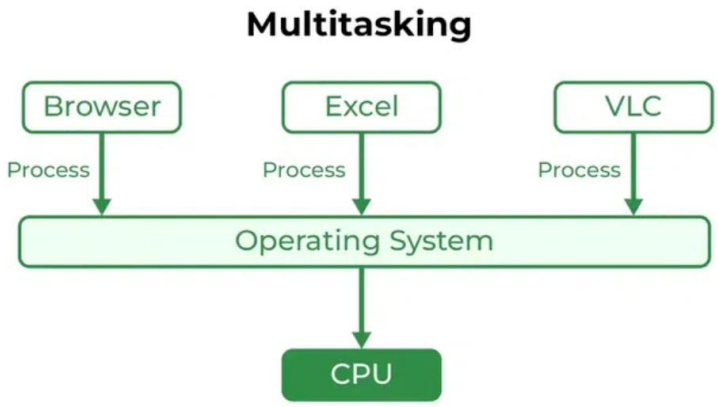
- Due to the multiple CPU, it can be more complex and somehow difficult to understand.

4. Multi-Tasking Operating System

Multitasking Operating System is simply a multiprogramming Operating System with having facility of a Round-Robin Scheduling Algorithm. It can run multiple programs simultaneously.

There are two types of Multi-Tasking Systems which are listed below.

- Preemptive Multi-Tasking
- Cooperative Multi-Tasking



Advantages of Multi-Tasking Operating System

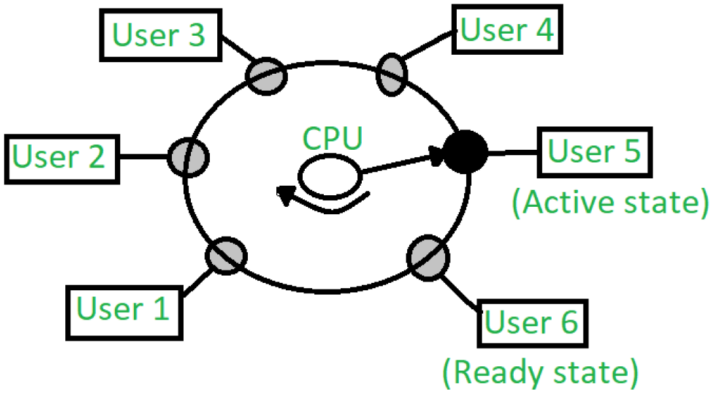
- Multiple Programs can be executed simultaneously in Multi-Tasking Operating System.
- It comes with proper memory management.

Disadvantages of Multi-Tasking Operating System

- The system gets heated in case of heavy programs multiple times.

5. Time-Sharing Operating Systems

Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of the CPU as they use a single system. These systems are also known as Multitasking Systems. The task can be from a single user or different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to the next task.



Advantages of Time-Sharing OS

- Each task gets an equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Resource Sharing: Time-sharing systems allow multiple users to share hardware resources such as the CPU, memory, and peripherals, reducing the cost of hardware and increasing efficiency.

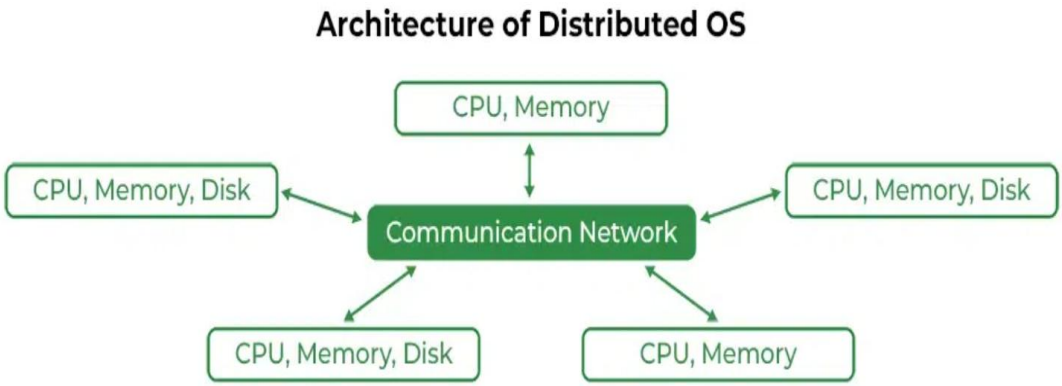
Disadvantages of Time-Sharing OS

- Reliability problem.
- One must have to take care of the security and integrity of user programs and data.
- Data communication problem.
- High Overhead: Time-sharing systems have a higher overhead than other operating systems due to the need for scheduling, context switching, and other overheads that come with supporting multiple users.
- Complexity: Time-sharing systems are complex and require advanced software to manage multiple users simultaneously. This complexity increases the chance of bugs and errors.
- Security Risks: With multiple users sharing resources, the risk of security breaches increases. Time-sharing systems require careful management of user access, authentication, and authorization to ensure the security of data and software.

5. **Distributed Operating System**

   These types of operating system is a recent advancement in the world of computer technology and are being widely accepted all over the world and, that too, at a great pace. Various autonomous interconnected computers communicate with each other using a shared communication network. Independent systems possess their own memory unit and CPU. These are referred to as loosely coupled systems or distributed systems. These

systems' processors differ in size and function. The major benefit of working with these types of the operating system is that it is always possible that one user can access the files or software which are not actually present on his system but some other system connected within this network i.e., remote access is enabled within the devices connected in that network.



Advantages of Distributed Operating System

- Failure of one will not affect the other network communication, as all systems are independent of each other.
- Electronic mail increases the data exchange speed.
- Since resources are being shared, computation is highly fast and durable.
- Load on host computer reduces.

Disadvantages of Distributed Operating System

- Failure of the main network will stop the entire communication.
- To establish distributed systems the language is used not well-defined yet.
- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.

6. Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called response time.

Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Types of Real-Time Operating Systems

- **Hard Real-Time Systems:**
  Hard Real-Time OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of an accident. Virtual memory is rarely found in these systems.
- **Soft Real-Time Systems:**
  These OSs are for applications where time-constraint is less strict.

2. Operating systems for personal computers & workstations

Operating systems for personal computers and workstations are designed to provide a user-friendly interface and support a wide range of applications for everyday computing tasks and productivity. Here are some common operating systems used in personal computers and workstations:

1. Windows 10: Windows 10, developed by Microsoft, is one of the most widely used operating systems for personal computers and workstations. It offers a familiar graphical user interface (GUI) and supports a vast array of software applications. Windows 10 is known for its compatibility with a wide range of hardware and peripherals, making it a popular choice for many users.

2. macOS: macOS, developed by Apple Inc., is the operating system used on Apple Macintosh computers. It provides an intuitive and user-friendly interface, along with seamless integration with other Apple devices such as iPhones and iPads. macOS is favored by many creative professionals due to its robust multimedia capabilities and software ecosystem.

3. Linux: Linux is an open-source operating system that offers a high degree of flexibility and customization. It comes in various distributions (distros), each tailored to specific user needs. Ubuntu, Fedora, and Debian are some popular desktop Linux distributions that provide a user-friendly experience, extensive software repositories, and excellent hardware support.

4. Chrome OS: Chrome OS is a lightweight operating system developed by Google, primarily designed for use with Chromebooks and Chromebox devices. It is cloud-centric, relying on web-based applications and Google's suite of productivity tools. Chrome OS is known for its fast boot times and low resource requirements, making it suitable for simple computing tasks and web browsing.

5. FreeBSD: FreeBSD is an open-source Unix-like operating system that offers high performance, stability, and robust security features. While it is primarily used in server environments, it is also a viable choice for power users and developers seeking a Unix-like experience on their personal computers or workstations.
6. Haiku: Haiku is an open-source operating system inspired by the classic BeOS. It is designed for modern hardware and offers a unique user interface and features like native support for media handling and multiprocessing. Haiku is still in development but showcases promise for users looking for a different computing experience.
7. ReactOS: ReactOS is an open-source operating system aiming to be binary-compatible with Microsoft Windows. It allows users to run Windows applications without the need for a Microsoft Windows license. Although still in development, ReactOS shows potential as a free alternative for Windows users.

Each of these operating systems has its strengths and unique features, making them suitable choices for different user preferences, needs, and computing environments. The choice between these operating systems depends on factors such as hardware compatibility, software requirements, user familiarity, and desired level of customization and control.

3. Process control and real-time systems

1. Process Control: Process control refers to the use of computer systems and algorithms to monitor and regulate industrial processes, machines, or systems in real-time. The main objective of process control is to maintain the desired output or performance by adjusting various parameters and variables based on feedback from sensors and other monitoring devices. This feedback loop allows the system to react to changes and disturbances and maintain stability and efficiency.

Process control systems are commonly used in industries such as manufacturing, chemical processing, power generation, water treatment, and more. The components of a process control system typically include sensors to measure process variables (e.g., temperature, pressure, flow rate), a controller to process the sensor data and generate control signals, and actuators to adjust the process variables accordingly (e.g., adjusting valves, motors, or pumps).

Process control can be broadly categorized into two types:

- Open-loop Control: In open-loop control, the control action is determined based on a predefined setpoint, without considering the system's output or feedback. It is less

common in industrial applications and is mostly used in simple systems with no feedback mechanisms.

- Closed-loop Control (Feedback Control): In closed-loop control, the control action is continuously adjusted based on feedback from the system's output. This allows the system to respond to disturbances and maintain the desired output. Most industrial process control systems use closed-loop control.

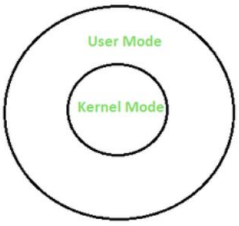
2.Real-Time Systems: Real-time systems are computer systems or applications that are designed to respond to external events or inputs within strict time constraints. In real-time systems, the correctness of the system's behavior not only depends on the output's accuracy but also on the time at which the output is produced. The primary goal of real-time systems is to ensure timely and predictable responses.

Real-time systems can be classified into two main categories:

- Hard Real-Time Systems: In hard real-time systems, meeting the specified deadlines is crucial. If a deadline is missed, it can lead to catastrophic consequences. These systems are commonly found in safety-critical applications like aircraft control systems, medical devices, and automotive systems.
- Soft Real-Time Systems: Soft real-time systems have less stringent timing requirements. While meeting deadlines is important, occasional missed deadlines may not cause catastrophic failures. Examples of soft real-time systems include multimedia streaming, online gaming, and video conferencing.

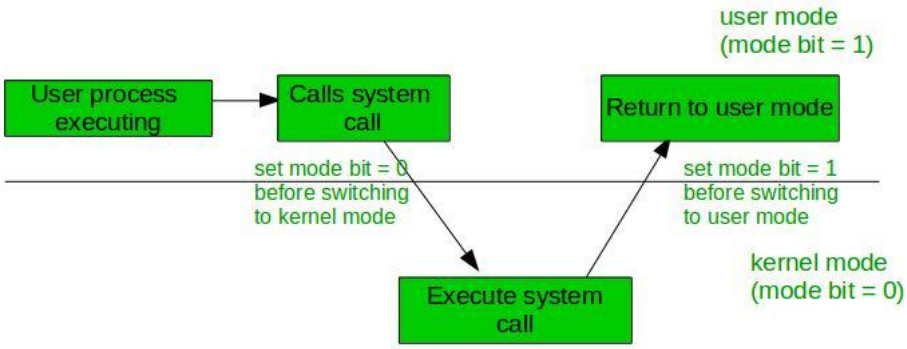
4.User Mode & Kernel Mode:

User Mode: When a Program is booted up on an Operating system let's say windows, then it launches the program in user mode. And when a user-mode program requests to run, a process and virtual address space (address space for that process) is created for it by windows.



User-mode programs are less node applications and are not allowed to access the system resources directly. For instance, if an application under user-mode wants to access system resources, it will have to first go through the Operating system kernel by using syscalls.

Kernel Mode: The kernel is the core program on which all the other operating system components rely, it is used to access the hardware components and schedule which processes should run on a computer system and when, and it also manages the application software and hardware interaction. Hence it is the most privileged program, unlike other programs it can directly interact with the hardware. When programs running under user mode need hardware access for example webcam, then first it has to go through the kernel by using a syscall, and to carry out these requests the CPU switches from user mode to kernel mode at the time of execution. After finally completing the execution of the process the CPU again switches back to the user mode.



Difference Between Kernel mode and User mode:

Kernel Mode	User Mode
In kernel mode, the program has direct and unrestricted access to system resources.	In user mode, the application program do nat have direct access to system resources. In order to access the resources, a system call must be made.
In kernel mode, the program has direct and unrestricted access to system resources.	In user mode, a single process fails if an interrupt occurs.
Kernel mode is also known as the master mode, privileged mode, or system mode.	User mode is also known as the unprivileged mode, restricted mode, or slave mode.
kernel mode, all processes share a single virtual address space.	In user mode, all processes get separate virtual address space

Kernels, system calls, and system programs:

Kernel: The kernel is the core of an operating system. It is a low-level software that manages the system's resources and provides an interface between the hardware and the higher-level software (applications and system programs). The main functions of the kernel include memory management, process management, device management, and handling system interrupts. It controls the allocation of hardware resources to different processes, ensures their isolation and protection, and facilitates communication between different parts of the system.

System Calls: System calls are the interface through which user-level applications interact with the kernel. When a user program running in user mode requires access to privileged operations (e.g., reading/writing files, creating a new process, allocating memory), it must request the kernel to perform those operations on its behalf. The user program invokes system calls by making a request to the operating system using a specified API (Application Programming Interface). The system call acts as a gateway to the kernel's functionality, allowing user programs to perform tasks that require elevated privileges.

System Programs: System programs, also known as utility programs or system utilities, are a collection of software applications that come bundled with the operating system. These programs perform various essential tasks and provide a convenient way for users to interact with the system and perform common operations. System programs include text editors, file management utilities, device drivers, command-line interpreters (shell), network utilities, and more. They utilize system calls to access kernel services and perform tasks that may not be directly achievable through regular user-level applications.

MCQ:

SL. No.	Question	Option A	Option B	Option C	Option D	Correct Answer
1	What is an operating system?	A program that manages computer hardware	A set of computer hardware components	A collection of software applications	A user interface for a computer	Option A

SL. No.	Question	Option A	Option B	Option C	Option D	Correct Answer
2	Which of the following is NOT an operating system?	Windows 10	Microsoft Office	macOS	Linux	Option B
3	What is the primary purpose of a batch processing operating system?	To provide a user-friendly interface	To optimize CPU utilization	To ensure tasks meet strict timing requirements	To manage resources efficiently	Option D
4	Select the example of a time-sharing operating system	Windows	Linux	macOS	Unix	Option D
5	What is the primary purpose of an operating system?	To allow users to play games	To manage computer hardware and software resources	To create documents	To connect to the internet	Option B
6	Which type of operating system allows multiple tasks to be loaded into memory simultaneously?	Batch Processing	Multiprogramming	Time-Sharing	Real-Time	Option B
7	What is the role of the kernel in an operating system?	To provide a user interface	To manage computer resources	To run applications	To store files and folders	Option B

SL. No.	Question	Option A	Option B	Option C	Option D	Correct Answer
8	Which scheduling algorithm is used to allocate CPU time fairly?	Round Robin	First Come, First Serve	Shortest Job Next	Priority Scheduling	Option A
9	What is the main advantage of a multiprogramming operating system?	Improved CPU utilization	Real-time task scheduling	Enhanced user interactivity	Better support for large-scale computations	Option A
10	Which type of operating system is most suitable for applications that require immediate and continuous response?	Batch Processing	Multiprogramming	Time-Sharing	Real-Time	Option C

Short Answer Questions:

1. **What is an operating system?**
2. **What are the primary functions of an operating system?**
3. **What is the role of the kernel in an operating system?**
4. **What is the purpose of memory management in an operating system?**
5. **What is multitasking in an operating system?**
6. **Describe distributed operating systems.**
7. **What is the difference between a real-time operating system and a general-purpose operating system?**
8. **Explain the concept of time-sharing in an operating system.**
9. **What is virtual memory, and why is it used?**
10. **What is a file system, and what does it do?**

Long Answer Questions:

1. **What is Operating System? Explain the different types of operating system.**
2. **Describe structure of Operating System.**
3. **Explain the major components or modules of an operating system, and what are their roles.**
4. **Describe User Mode & Kernel Mode of operating system.**
5. **Describe Distributed Operating System with example.**

Advanced Learner Questions:

Case Study 1: Managing Memory in a Multi-Programmed System

Scenario: A company uses a multi-programmed operating system to run multiple applications simultaneously on their server. The server has limited physical memory, and the company needs to manage memory efficiently to ensure optimal performance and prevent system slowdowns.

Questions:

1. **Describe the basic function of memory management in an operating system. How does the operating system allocate and manage memory for different processes?**
2. **Explain the concept of virtual memory and its role in memory management. How does virtual memory allow the system to use secondary storage (e.g., hard disk) as an extension of physical memory?**
3. **Discuss the potential challenges of memory fragmentation in a multi-programmed system. How can the operating system address fragmentation to maintain efficient memory utilization?**
4. **The company experiences slowdowns when multiple memory-intensive applications run concurrently. Suggest strategies the operating system can employ to improve memory performance in such scenarios.**

Suggested Books:

1. A Silberschatz, P.B. Galvin, G. Gagne, Operating Systems Concepts, 8th Edition, John Wiley Publications 2008.

2. A.S. Tanenbaum, Modern Operating Systems, 3rd Edition, Pearson Education 2007.
3. G. Nutt, Operating Systems: A Modern Perspective, 2nd Edition Pearson Education 1997.
4. W. Stallings, Operating Systems, Internals & Design Principles , 5th Edition, Prentice Hall of India. 2008.
5. M. Milenkovic, Operating Systems- Concepts and design, Tata McGraw Hill 1992.